Reconstructing 3-D Thermohaline Structures for Mesoscale Eddies Using Satellite Observations and Deep Learning

Yingjie Liu, *Member, IEEE*, Haoyu Wang, *Member, IEEE*, Fei Jiang, *Graduate Student Member, IEEE*, Yuan Zhou, *Senior Member, IEEE*, and Xiaofeng Li, *Fellow, IEEE*

***Abstract*— Mesoscale eddies are circular water currents found widely in the ocean and significantly impact the ocean's circulation, water distribution, and biology. However, our comprehension of eddies' 3-D structures remains constrained due to the scarcity of in situ data. Therefore, we introduce a novel deep learning (DL) model, 3D-EddyNet, designed for reconstructing the 3-D thermohaline structure of mesoscale eddies. Utilizing multisource satellite data and Argo profiles collected from eddies in the North Pacific Ocean between 2000 and 2015, we optimized the 3D-EddyNet model by adjusting image sizes, introducing a convolutional block attention module (CBAM), and incorporating eddy physical parameters. The results demonstrate remarkable accuracy, with an average root mean square error (RMSE) of 0.32 °C (0.03 psu) for temperature (salinity) within anticyclonic eddies and 0.41 °C (0.04 psu) within cyclonic eddies in the upper 1000 m. We applied 3D-EddyNet to reconstruct 3-D eddy structures in the Kuroshio extension (KE) and the Oyashio current (OC) regions, demonstrating its capability to accurately represent the 3-D thermohaline eddy structures both vertically and horizontally. The consistency in the averaged 3-D eddy structures between our 3D-EddyNet and the ARMOR3D dataset in the KE and OC regions underscores the robust generalizability of our model, indicating the model's ability to infer 3-D eddy structures when Argo profiles are unavailable. The distinctive advantage offered by 3D-EddyNet enhances our ability to understand mesoscale eddy dynamics, overcoming challenges posed by the limited availability of in situ data.**

***Index Terms*— Deep learning (DL), mesoscale eddies, prior knowledge-embedded, reconstruction of 3-D thermohaline structure.**

I. Introduction

MESOSCALE eddies are ubiquitous in the ocean, characterized by time scales ranging from several days to several hundred days and horizontal scales spanning from tens to hundreds of kilometers. They can also extend vertically from tens to thousands of meters and play a critical role in transporting and mixing oceanic matter and energy [1], [2], [3], [4], [5]. On the one hand, mesoscale eddies facilitate the movement of water masses with distinct temperature, salinity, biological, and chemical properties [1], [5], [6], [7]. Their global impact can be likened to large-scale wind-driven and thermohaline circulation systems, significantly influencing the distribution of energy, heat, matter, and freshwater throughout the world's oceans [5], [7]. On the other hand, the vertical process inside mesoscale eddies can bring cold water and nutrients from below the sea surface to the sea surface or warm water from the sea surface to deep layers. It leads to geostrophic depression [for an anticyclonic eddy (AE)] and uplift [for a cyclonic eddy (CE)] of the background pycnocline, affecting the vertical structure of the ocean's temperature and salinity and the distribution of primary productivity [1], [8]. Consequently, gaining insights into mesoscale eddies' 3-D structure is imperative for comprehending global oceanic mass and energy transport, oceanic dynamic processes, and marine ecosystems.

With the accumulation of satellite remote sensing data, including microwave, infrared, and visible light imagery, extensive analysis has benefitted from long-term satellite observations. These observations provide a synoptic view of the surface manifestation of mesoscale features and eddies, greatly enhancing our understanding of the global distribution, kinematic attributes, and spatiotemporal evolution of mesoscale eddies [2], [9], [10], [11], [12]. However, satellite observations cannot obtain the subsurface structure of mesoscale eddies. The emergence of various in situ observation techniques, such as conductivity, temperature, depth (CTD), eXpendable BathyThermograph (XBT), acoustic Doppler current profilers (ADCPs), gliders, and mooring arrays, has enabled researchers to probe the vertical structure of mesoscale eddies at deeper depths [13], [14], [15], [16], [17]. For instance, Garreau et al. [13] monitored the vertical structure of a specific AE in the Algerian Basin from its generation to its dissipation using high-resolution hydrological transects, CTD casts, and lowered acoustic Doppler current profiler (LADCP) measurements. Shu et al. [17] conducted field observations using 12 gliders and 62 expendable probes to investigate the 3-D structure and temporal evolution of an AE in the northern South China Sea during the Summer of

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Yingjie Liu, Haoyu Wang, Fei Jiang, and Xiaofeng Li are with the Key Laboratory of Ocean Circulation and Waves, Institute of Oceanology, Chinese Academy of Sciences, Qingdao 266071, China (e-mail: yjliu@qdio.ac.cn; wanghy@qdio.ac.cn; jiangfei@qdio.ac.cn; xiaofeng.li@ieee.org).

Yuan Zhou is with the School of Electrical and Information Engineering, Tianjin University, Tianjin 300072, China (e-mail: zhouyuan@tju.edu.cn).

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2017. However, previous studies primarily focused on case studies of subsurface thermohaline structures within mesoscale eddies.

Since the 21st century, the deployment of the Array for Real-time Geostrophic Oceanography (Argo) floats in the global ocean has expanded significantly, becoming a major component of the ocean observation system. Argo observations are primarily employed to study the composite 3-D structure of mesoscale eddies in the global ocean and various oceanic regions. For instance, Zhang et al. [3] observed that mesoscale eddies' horizontal and vertical structures can be distinguished and exhibit a unified spatial structure under the geostrophic approximation in the global ocean. In the northwestern subtropical Pacific Ocean, Yang et al. [18] discerned that the temperature anomalies induced by mesoscale eddies have a dual-core vertical structure, while salinity anomalies assume a "sandwich" shape. In the southeastern tropical Indian Ocean, Yang et al. [19] determined that ocean temperature and salinity anomalies caused by mesoscale eddies are predominantly concentrated above 300 m. In the Southern Ocean, Frenger et al. [20] discovered that the mesoscale eddy signals can extend to depths of 2000 m. In the Kuroshio extension (KE) region, Sun et al. [21] identified bullet-shaped mesoscale eddies with temperature and salinity anomalies primarily limited to depths of 800 m. In the South China Sea, He et al. [22] observed long conical-shaped mesoscale eddies, with the composite temperature and salinity anomalies within CEs (AEs) being most pronounced at depths of 60 m (90 m).

Previous studies have revealed the global consistency of mesoscale eddies' 3-D structure and regional variations using Argo profiles. It is paramount for evaluating eddies' impact on oceanic mass and energy transport. However, despite the collection of over 2 million Argo profiles worldwide since 2000, not all profiles are suitable for analyzing the 3-D structure of mesoscale eddies [23]. The spatiotemporal resolution of Argo profiles within eddies still lags behind remote sensing data. Consequently, most previous studies estimate the 3-D structure of mesoscale eddies through a composite analysis method, which provides a mean representation of the 3-D structure of mesoscale eddies. However, this approach overlooks individual variations and spatiotemporal fluctuations in the 3-D structure of mesoscale eddies throughout their lifecycles, resulting in uncertainties in eddy transport estimation [5]. Therefore, finding solutions to overcome the constraints posed by the sparse distribution of in situ data and to obtain high-precision, spatiotemporally continuous 3-D eddy structures is an urgent challenge.

The surface and interior ocean exhibit a strong nonlinear relationship and are highly correlated [24]. Three methods have been proposed to obtain continuous 3-D ocean state information, leveraging high temporal and spatial resolution remote sensing observations to map surface features to subsurface layers accurately. These methods include physics-driven dynamic method, empirical statistical method, and artificial intelligence (AI)-based method. The dynamic method employs a dynamic model to convert surface data into control variables for the model, thereby achieving mapping from the surface layer to the subsurface layer. For instance, Haines [25] utilized a four-layer quasi-geostrophic model to reconstruct the subsurface flow field by the surface flow represented by sea surface height (SSH). Wang et al. [26] proposed the "interior+surface quasi-geostrophy equation" (isQG) dynamic method, which fully utilizes surface data such as SSH, sea surface temperature (SST), sea surface salinity (SSS), and historical in situ profiles to reconstruct subsurface velocity and motion [27]. The dynamic method, exemplified by isQG, adheres to the physical framework and offers a significantly lower computational cost than ocean numerical models. Nevertheless, their reliance on simplified physical assumptions and the incomplete dynamic framework may result in a lack of constraints among elements [28]. With the improvement of remote sensing data's resolution and spatial coverage, some studies have adopted empirical statistical models, such as the least-square regression method and the multiple linear regression method, to retrieve subsurface temperature and salinity fields in ocean basins using satellite SST and SSH data [29], [30], [31]. However, the empirical statistical method lacks physical constraints and primarily relies on linear assumptions, making it challenging to establish stable statistical relationships in areas with limited observation profiles or significant changes.

In contrast to empirical statistical methods, AI technology demonstrates a strong ability to learn and model complex relationships, which has recently been used in intelligent analysis and mining ocean big data [32], [33], [34], [35], [36], [37], [38], [39]. A significant amount of literature has successfully employed AI-based methods, such as self-organizing maps, support vector machines, random forests, deep neural networks (DNNs), and convolutional neural networks (CNNs), for inverting subsurface temperature and salinity fields from satellite observations [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50]. However, the majority of previous studies have primarily focused on large-scale ocean basins, with fewer investigations into the inversion of subsurface temperature and a notable absence of studies addressing the inversion of subsurface salinity in mesoscale eddies.

Several challenges persist before achieving high-precision, spatiotemporally continuous 3-D thermohaline structures of mesoscale eddies. On the one hand, existing studies primarily involve inverting the subsurface temperature at specific points within eddies. Still, they lack the inversion of horizontal changes in thermohaline at different depth layers within eddies, preventing the revelation of the complete 3-D thermohaline structure of eddies. On the other hand, in contrast to physics-driven dynamic methods, purely data-driven AI-based approaches lack constraints from physical knowledge, posing challenges in ensuring alignment with established physical laws during the 3-D thermohaline inversion of eddies.

To address these challenges, the study proposes a deep learning (DL)-based model, namely, 3D-EddyNet. It is designed to reconstruct the 3-D temperature and salinity structures of eddies by incorporating physical prior knowledge of eddies, multisource remote sensing data, and Argo profiles (see Fig. 1). Section II provides an overview of the data and study area, while Section III outlines the methodology. The performance evaluation of the 3D-EddyNet model is detailed

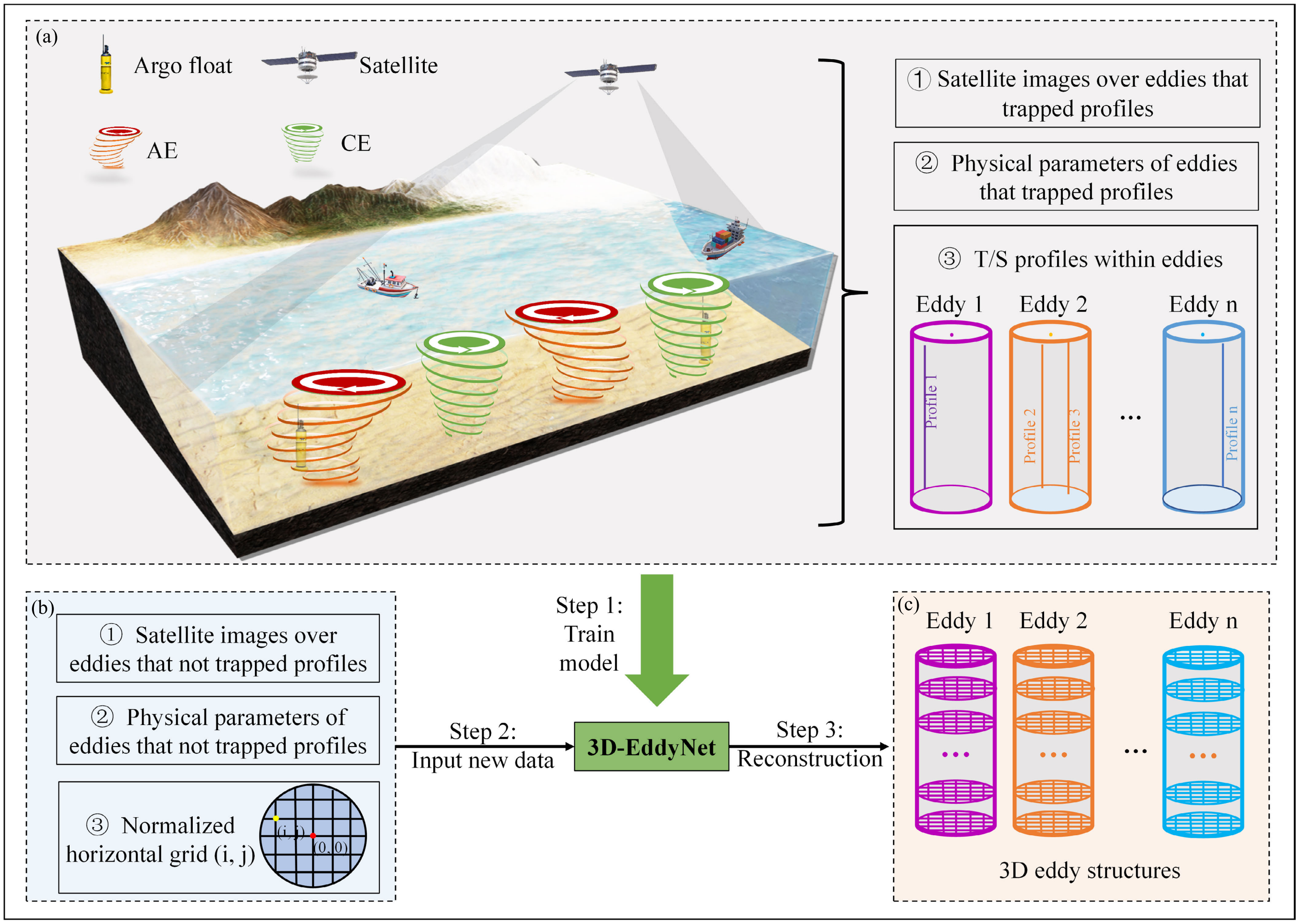

Fig. 1. Flowchart of the reconstruction of 3-D eddy structure using the 3D-EddyNet model. Cylinders are employed to approximate the envelope of the 3-D eddy structure.

in Section IV, with specific results presented in Section V. Finally, Section VI offers discussion and conclusions.

II. Data

A. Multisource Satellite Data

The study utilizes four remote sensing products: SSH, SST, SSS, and sea surface wind (SSW) data.

1) The SSH product, available daily at a spatial resolution of 0.25°, was provided by the European Copernicus Marine Environment Monitoring Service (CMEMS). The product was merged from all available altimeter missions, including TOPEX/Poseidon (TP), Jason-1&2, European Remote-Sensing Satellite (ERS)-1&2, Environmental Satellite (ENVISAT), Geosat Follow On (GFO), Cryosat-2, Saral/AltiKa, and Haiyang-2A.

2) The SST dataset is the NOAA Optimum Interpolation (OI) SST product from [51] on daily and 0.25° resolution. The OISST dataset was constructed from infrared satellite observations of the Advanced Very High Resolution Radiometer (AVHRR) with supplemental information provided by in situ observations and proxy SSTs computed from sea ice concentrations.

3) The SSS product is available weekly at a resolution of 0.25° and was provided by CMEMS. This product was obtained through a multivariate optimal interpolation algorithm that combines SSS images from various satellite sources, such as NASA's Soil Moisture Active Passive (SMAP) and ESA's Soil Moisture Ocean Salinity (SMOS) satellites, with in situ salinity measurements and satellite SST information.

4) The SSW data were obtained from the Cross-Calibrated Multi-Platform (CCMP) wind dataset, which was produced using satellite, moored buoy, and model wind data provided by Remote Sensing Systems (RSS). The SSW product includes U-wind at 10 m (SSWU) and V-wind at 10 m (SSWV), which are available on 6 h and 0.25° spatial resolution.

The primary aim of this study is to reconstruct the 3-D temperature and salinity structures of mesoscale eddies at any given date and location. To achieve this objective, remote sensing products have been standardized to a daily average temporal resolution. To maintain consistent temporal alignment with daily SSH and SST data, we employed the linear interpolation method to derive daily averaged SSS data from weekly measurements. Furthermore, daily SSW data were computed by averaging the 6-h SSW measurements within the same day.

To identify eddy-induced surface anomalies, we applied temporal and spatial filters similar to those used by Bôas et al. [52] on the fields of SSH, SST, SSS, SSWU, and

SSWV. The temporal filter is a bandpass Butterworth window, designed to retain signals with periods between 7 and 90 days, while the spatial filter is a moving average Hann window, preserving spatial scales smaller than 600 km. In addition, we employed normalization processing on each remote sensing dataset due to the variations in units and value ranges among different surface parameters. This process effectively reduces the magnitude of the values while preserving crucial information, ultimately enhancing the accuracy of our model inversion.

B. Satellite-Derived Mesoscale Eddy Dataset

The mesoscale eddy dataset was generated by a DL-based model [53] using satellite-derived SSH anomaly (SSHA) and SST anomaly (SSTA) data. The eddy dataset covers daily observations across the global ocean from 1993 to 2015. In this study, we employ fundamental physical parameters of mesoscale eddies as prior knowledge for the 3D-EddyNet model. The selected parameters—eddy radius, amplitude, velocity (U), and eddy kinetic energy (EKE)—are chosen for their efficacy in capturing and reflecting the intrinsic physics that govern the behavior of eddies [2], [11], [12], [54].

Eddy radius is the radius of a circle with the same area as the region within the eddy boundary, indicating the size of the eddy structure in the horizontal direction. Eddy amplitude is the SSHA difference between the eddy center and the eddy boundary, representing the strength of the eddy. Eddy velocity reflects the rotational speed of the eddy, which can be calculated as

$$U = \sqrt{u^2 + v^2}. \tag{1}$$

Here, u and v are geostrophic velocity components that can be computed from the SSHA gradients

$$u = -\frac{g}{f}\frac{\partial(\mathrm{SSHA})}{\partial y} \tag{2}$$

$$v = \frac{g}{f}\frac{\partial(\mathrm{SSHA})}{\partial x} \tag{3}$$

where g is the acceleration due to gravity; f is the Coriolis parameter; and ∂x and ∂y are the eastward and northward distances, respectively. Moreover, EKE is a key metric commonly used to measure the intensity and activity of eddies, which can be calculated as

$$\mathrm{EKE} = \frac{u^2 + v^2}{2}. \tag{4}$$

Considering the resolution and precision of the SSHA product [55], we selected individual eddies with amplitudes $\geq$1 cm and radii $\geq$35 km for the study.

C. In Situ Data

In the study, Argo data are employed as the ground-truth dataset for reconstructing the 3-D structure of eddies. The Argo data are provided by the Coriolis Global Data Acquisition Center of France, including temperature and salinity measurements spanning from the sea surface to thousands of meters below. Only Argo profiles co-located within 1.5 eddy radii (R) are considered in the study. Quality control was first applied to eddy-collocated Argo profiles: 1) only temperature, salinity, and pressure data with quality flags 1 (good data) and 2 (probably good data) are considered; 2) each profile must include a shallow data point between 0 and 10 m and a deep data point below 1000 m; and 3) the number of observation samples between 10 and 1000 m should be larger than 30. Second, the temperature/salinity profiles are interpolated on a regular 10 m from 10 to 1000 m. Third, the daily climatological temperature and salinity values provided by the CSIRO Atlas of Regional Sea 2009 (CARS2009) dataset [56], [57] are subtracted from Argo profiles to obtain anomalies in temperature and salinity within eddies. The CARS2009 data were based on all available marine observation data and automatic buoy profile data in history, which had passed the quality control and were gridded on a grid of 0.5° × 0.5°. Compared with other climatology data, CARS2009 is more suitable for the western boundary region [18].

D. ARMOR3D Data

To further validate the accuracy of the 3-D eddy structures inverted by the 3D-EddyNet model, we utilize the ARMOR3D data [58], [59], a 3-D thermohaline product widely used to delineate the vertical structure of mesoscale eddies in the various regions [60], [61], [62], [63], [64], [65], [66], [67]. The ARMOR3D product, launched by CMEMS, combines satellite estimates of the SSHA, SST, and mean dynamic topography (MDT) with in situ observations using regression and optimal interpolation. The ARMOR3D dataset provides globally weekly 3-D combined temperature and salinity fields on a 0.25° horizontal grid with 50 unevenly spaced layers spanning from the surface to a depth of 5500 m. In the study, we focus on the top 36 layers (0–1000 m) of the ARMOR3D dataset to align with the depth range of Argo profiles. To obtain eddy-induced temperature and salinity anomalies, we subtract monthly climatology from the ARMOR3D data

$$\overline{P}_M(z, j, i) = \frac{1}{N}\sum_{n=1}^{N} P_{w\in M}(z, j, i) \tag{5}$$

$$P'_w(z, j, i) = P_w(z, j, i) - \overline{P}_{w\in M}(z, j, i) \tag{6}$$

where P denotes seawater temperature or salinity, Mrepresents the month, and (z, j, i) are the dimensions of depth, latitude, and longitude, respectively. N is the total number of weeks in a particular month during the study period. $\overline{P}_M$ is the monthly climatology of temperature or salinity, P_w is the original temperature or salinity value of ARMOR3D dataset, and P'_w is the temperature or salinity anomalies.

Subsequently, we extract the 3-D temperature and salinity fields for each eddy within a 2° × 2° box centered on the eddy's center. In this step, individual eddies are matched with the weekly 3-D temperature/salinity anomalies on the nearest date. Using the matched results, we construct 3-D structures of eddy-induced temperature and salinity anomalies through spatial and temporal averaging.

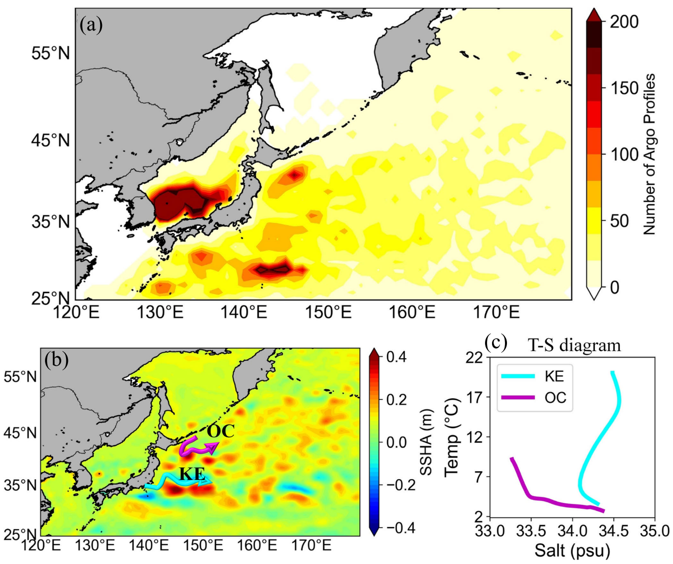

Fig. 2. (a) Spatial distribution of Argo profiles' number in the North Pacific Ocean region during 2000–2015. (b) Schematic of the KE and OC on the annual mean SSHA map, and (c) their T-S diagrams calculated based on Argo profiles.

E. Study Area

Our study primarily focuses on reconstructing 3-D eddy structures in the North Pacific Ocean (25°–60° N, 120°–180° E). As depicted in Fig. 2, this region encompasses two distinct currents characterized by differing thermohaline properties: the KE current, represented by the cyan line, known for its warm and saline waters, and the Oyashio current (OC), represented by the magenta line, recognized for its cold and fresh waters. Previous studies proposed that variations in background thermohaline fields lead to distinct 3-D thermohaline structures in eddies [23], [68]. Therefore, we can validate the model's robustness across different oceanic regions by comparing the 3-D eddy thermohaline structures inverted by 3D-EddyNet in the KE and OC regions.

III. Method

In this section, we designed the 3D-EddyNet model to reconstruct the 3-D thermohaline structure of mesoscale eddies. This model comprises an attention mechanism-based CNN branch and a prior knowledge-embedded DNN branch (see Fig. 3).

A. Attention Mechanism-Based CNN Branch

The CNN branch, which utilized an attention mechanism, took seven kinds of 2-D sea surface images of mesoscale eddies as input [see Fig. 3(a)]. It includes the latitude, longitude, SSHA, SSTA, SSSA, SSWUA, and SSWVA maps centered around eddy centers. The branch initiates with a sequence of convolutional layers with an increasing number of output channels: 64, 128, 256, and 512 channels. The filter size for all convolution layers is 3 × 3, with a stride of 1 and zero padding. Following the convolutional layers, a batch normalization (BN) layer is incorporated to stabilize the training process by normalizing activations in each channel. Subsequently, a convolutional block attention module (CBAM) [69] is applied to focus on both the "what" (channel) and "where" (spatial) aspects of the input. The CBAM enhances the feature extraction capability of the network by employing attention mechanisms in both channel and spatial dimensions. The CBAM's output then undergoes a max-pooling layer with a 3 × 3 window and a stride of 2, followed by a hyperbolic tangent activation function (Tanh). The pooling layer renders the outputs invariant and less sensitive to small translations in the input, reducing computational requirements.

B. Prior Knowledge-Embedded DNN Branch

The 3D-EddyNet not only extracted features from the sea surface images of eddies but also incorporated physical prior knowledge of eddies. Fig. 3(b) demonstrates the prior knowledge-embedded DNN branch, which takes 1-D physical eddy parameters as input, including EKE, eddy velocity, amplitude, and radius as input. Besides, locations of eddy centers (latitude and longitude) and the meridional and zonal distances (Δ latitude and Δ longitude) between Argo profiles and eddy centers are input into the model. It enables the 3D-EddyNet model to learn the relationship between eddy-induced temperature/salinity anomalies and their relative distances from eddy centers. In addition, we include the dates of mesoscale eddies to study temporal variations in eddy-induced temperature and salinity anomalies.

The prior knowledge-embedded DNN branch comprises three ResNet block modules, with 64, 128, and 256 neurons, respectively. Each ResNet block consists of three layers of neural networks with equal widths and two constant mappings. To facilitate information flow in the ResNet block, we use the "shortcut connection" method inspired by the residual module of ResNet. This method preserves the integrity of the input and minimizes information loss during propagation. When extending the network to a ResNet block in different channels, we include an additional neural network layer to match the output size with the input size of the next ResNet block.

Finally, we concatenate the outputs of both branches into a fully connected (FC) layer with 512 neurons. It is followed by two ResNet blocks with 256 and 128 neurons, and another FC layer with 128 neurons. This arrangement produces a scalar value representing subsurface temperature or salinity [see Fig. 3(c)].

C. Data Setting and Evaluation Metrics

In the study, we employed data from 2000 to 2015 for training, validation, and testing of the 3D-EddyNet model. In the North Pacific Ocean, 65 361 and 61 522 Argo profiles are located within 1.5 radii of AEs and CEs, respectively. For this study, we randomly allocate 60% of the sample database to the training set, 20% to the validation set, and 20% to the test set. We use the mean squared error (MSE) as the loss function to assess the 3D-EddyNet's performance. In addition, we employed various statistical metrics, including the coefficient of determination (R^2) and root mean square error (RMSE), to evaluate the model's ability to estimate

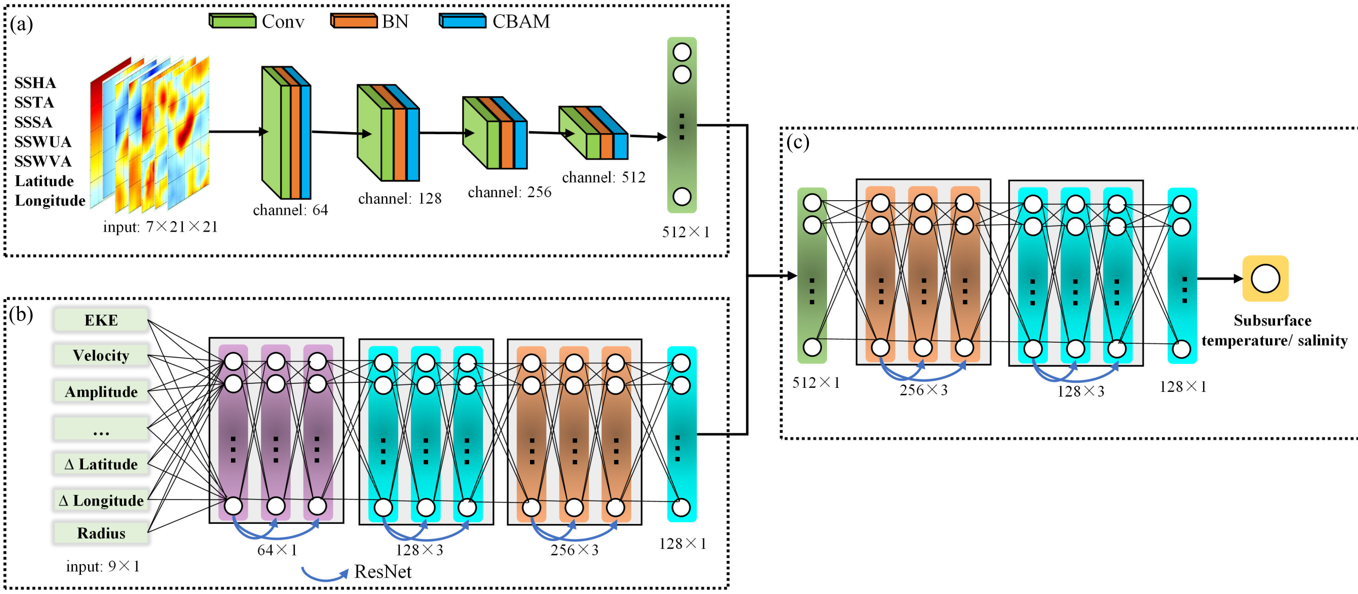

Fig. 3. Architecture of the 3D-EddyNet model. (a) Attention mechanism-based CNN branch. The input "7 × 21 × 21" denotes seven kinds of 2-D images each with size 21 × 21. (b) Prior knowledge-embedded DNN branch. The input "9 × 1" indicates nine kinds of 1-D eddy parameters. Δ Latitude (Δ Longitude) represents the meridional (zonal) distance between the Argo profile and the eddy center.

subsurface temperature and salinity. R^2 measures the proportion of variance in the dependent variable explained by the independent variables in a regression model, while RMSE quantifies the magnitude of errors between predicted outcomes and in situ data.

IV. Performance of the 3D-EddyNet Model

A. Model Setup

In the study, we conducted an ablation study, exploring the impact of modulating the dimensions (height × width) of 2-D sea surface images, introducing the CBAM module, and incorporating 1-D eddy physical parameters ($1D_{Phy}$) on the 3D-EddyNet model. Utilizing training and validation sets, we trained five models with distinct configurations.

1) *$1D_{Phy} + 2D_{CBAM}$ (11 × 11):* Simultaneous input of nine 1-D eddy parameters and seven 11 × 11-sized 2-D images into the 3D-EddyNet model incorporating CBAM.

2) *$1D_{Phy} + 2D_{CBAM}$ (31 × 31):* Simultaneous input of nine 1-D eddy parameters and seven 31 × 31-sized 2-D images into the 3D-EddyNet model incorporating CBAM.

3) *$1D_{Phy} + 2D_{CBAM}$ (21 × 21):* Simultaneous input of nine 1-D eddy parameters and seven 21 × 21-sized 2-D images into the 3D-EddyNet model incorporating CBAM.

4) *$1D_{Phy} + 2D$ (21 × 21):* Simultaneous input of nine 1-D eddy parameters and seven 21 × 21-sized 2-D images into the 3D-EddyNet model without CBAM.

5) *$2D_{CBAM}$ (21 × 21):* Input of only seven 21 × 21-sized 2-D images into the 3D-EddyNet model incorporating CBAM.

Following the training phase, we applied these models to the test dataset to evaluate their efficacy in inverting subsurface temperature and salinity within AEs and CEs, using R^2 and RMSE as performance metrics.

B. Evaluating the Optimal Input Size of Sea Surface Images

As shown in Fig. 4, R^2 (RMSE) values of $1D_{Phy} + 2D_{CBAM}$ (11 × 11) are markedly smaller (larger) compared to $1D_{Phy} + 2D_{CBAM}$ (21 × 21). Such a result suggests that larger sea surface images centered around eddy centers enhance the model's capacity to assimilate essential ocean surface background information related to eddies, thereby improving the accuracy of the 3D-EddyNet model. However, upon further expanding the size of sea surface images to 31 × 31, i.e., $1D_{Phy} + 2D_{CBAM}$ (31 × 31), it became evident that the R^2 and RMSE curves across various depths are similar to those observed with $1D_{Phy} + 2D_{CBAM}$ (21 × 21). The average values of R^2 (RMSE) curves remained consistent between the two models, as outlined in Table I. This observation suggests that enlarging sea surface images does not necessarily provide the model with additional insights into the surface background information of eddies. Importantly, using larger sea surface images also increases the model's training data volume. Therefore, carefully selecting an appropriate size of sea surface images is crucial for the model. Utilizing the mesoscale eddy dataset, we found the average radius of eddies in the North Pacific Ocean is approximately 250 km. Interestingly, the average size of eddies in this region aligns harmoniously with 21 × 21-sized sea surface images at a resolution of 0.25°. It can be indicated that models trained with sea surface images having an equal area to the average size of eddies in the study ocean region tend to exhibit superior performance in inverting subsurface structures of eddies.

C. Evaluating the Impact of CBAM

Next, we assessed the impact of removing the CBAM while keeping the size of sea surface images fixed at 21 × 21 for the 3D-EddyNet framework, denoted as $1D_{Phy} + 2D$ (21 × 21). As shown in Fig. 4, when the CBAM is excluded from the

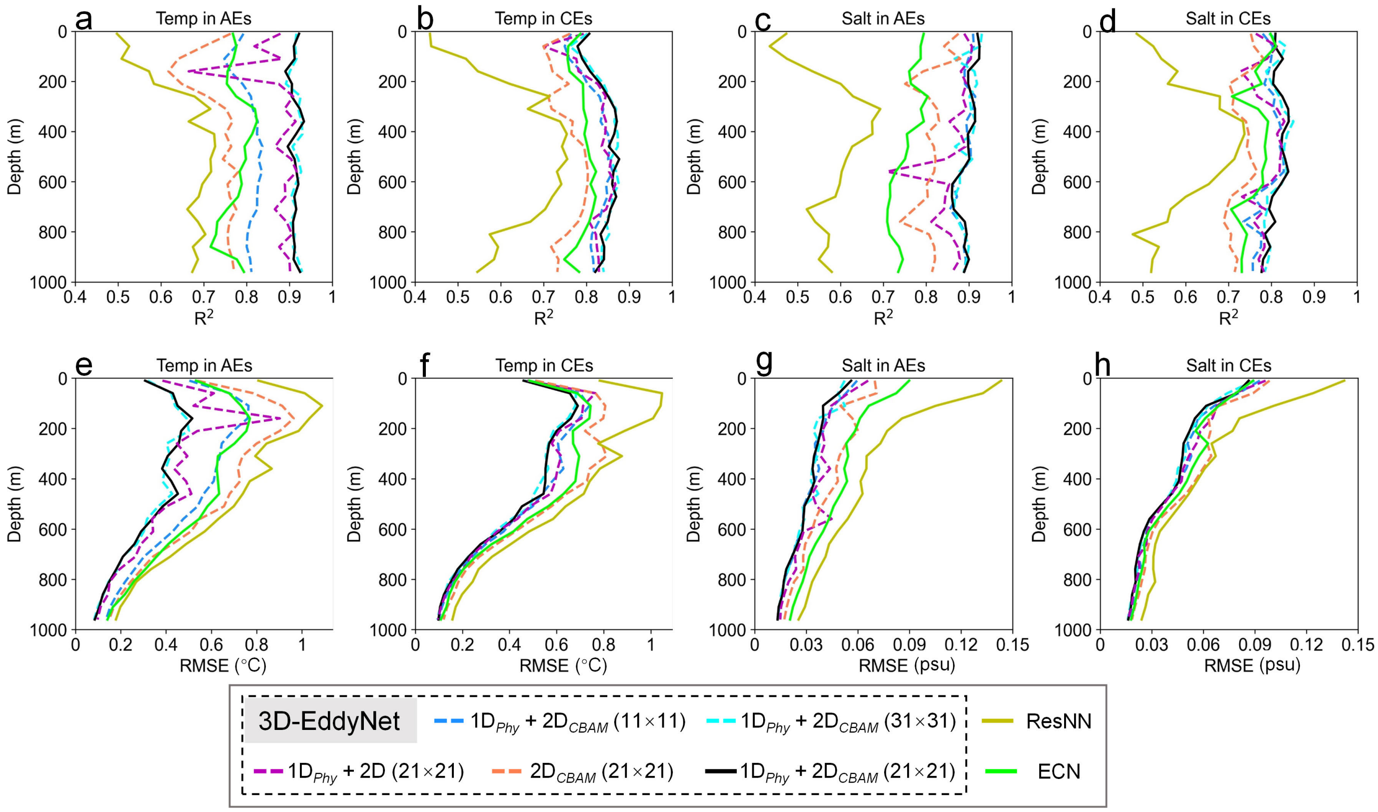

Fig. 4. R^2 (a)–(d) and RMSE (e)–(h) for the 3D-EddyNet with different configurations, ECN and ResNN validated by test dataset at different depth levels within AEs and CEs.

TABLE I
R^2 AND RMSE OF THE INVERSED TEMPERATURE AND SALINITY WITHIN AEs AND CEs

Model Type	Setup	R^2				RMSE			
		Temp		Salt		Temp		Salt	
		AEs	CEs	AEs	CEs	AEs	CEs	AEs	CEs
3D-EddyNet	$1D_{Phy}$ + $2D_{CBAM}$ (11×11)	0.81	0.82	0.89	0.79	0.48	0.44	0.03	0.04
	$1D_{Phy}$ + $2D_{CBAM}$ (31×31)	0.92	0.85	0.90	0.81	0.32	0.41	0.03	0.04
	$1D_{Phy}$ + 2D (21×21)	0.88	0.82	0.86	0.78	0.38	0.44	0.03	0.04
	$2D_{CBAM}$ (21×21)	0.74	0.76	0.81	0.72	0.57	0.51	0.04	0.05
	$1D_{Phy}$ + $2D_{CBAM}$ (21×21)	**0.92**	**0.85**	**0.90**	**0.81**	**0.32**	**0.41**	**0.03**	**0.04**
ECN	2D (21×21)	0.72	0.73	0.70	0.70	0.58	0.54	0.06	0.05
ResNN	$1D_{Phy}$ (14×1)	0.66	0.64	0.58	0.60	0.65	0.61	0.06	0.06

3D-EddyNet, the R^2 (RMSE) values for the inverted subsurface temperature and salinity within eddies displayed smaller (larger) values at various depths compared to models incorporating the CBAM. The result indicates the positive influence of incorporating the CBAM on enhancing the inversion performance of the 3D-EddyNet model. Furthermore, we found a significant deterioration in the model's inversion results at specific depths when the CBAM is removed. Specifically, the R^2 (RMSE) values for the subsurface temperature inverted by $1D_{Phy} + 2D$ (21×21) are smaller (larger) than that inverted by $1D_{Phy} + 2D_{CBAM}$ (21×21), particularly at approximately 200 m within AEs [see Fig. 4(a) and (e) and approximately

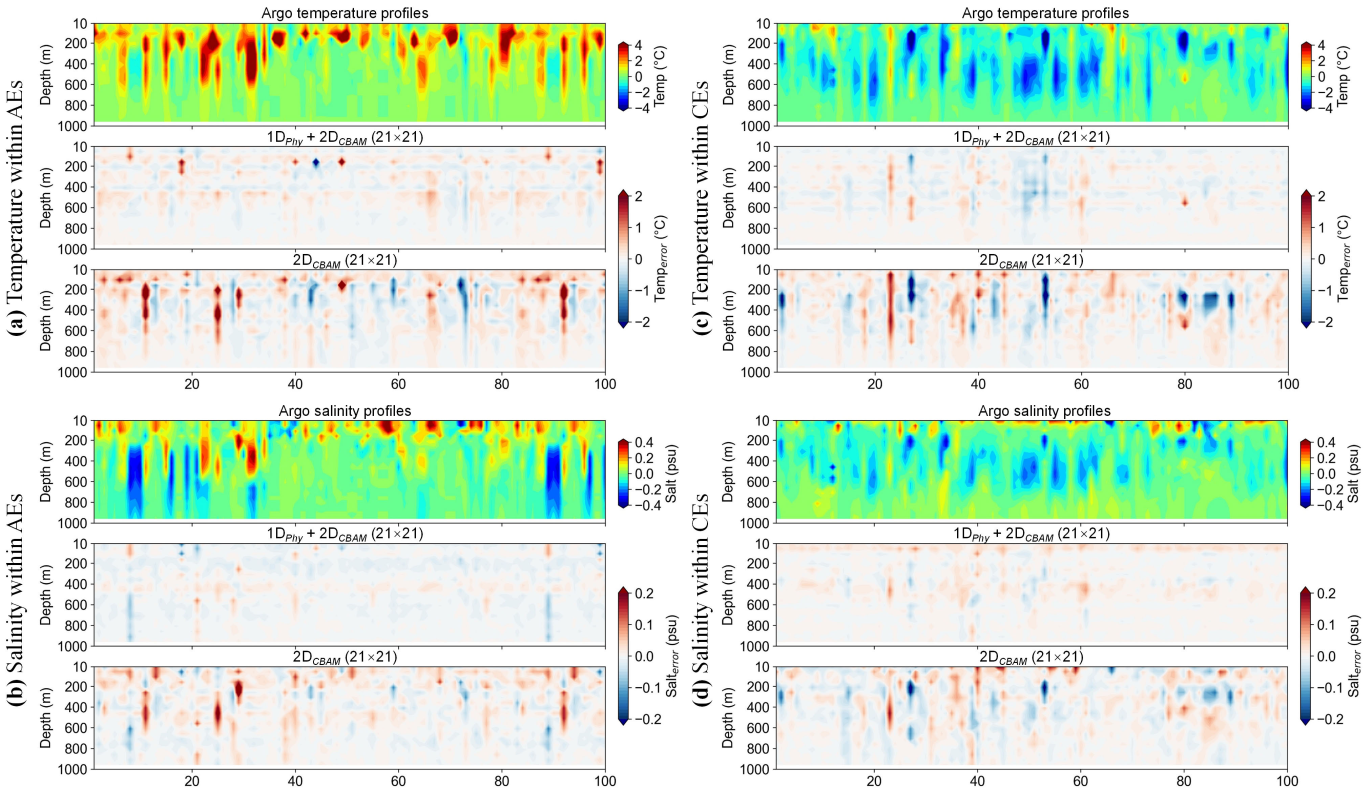

Fig. 5. Residual error of subsurface temperature and salinity within AEs and CEs inverted by $1D_{Phy}$ + $2D_{CBAM}$ (21 × 21) and $2D_{CBAM}$ (21 × 21). The residual error is obtained from ground truth minus the data inverted by the model. (a) Temperature within AEs. (b) Salinity within AEs. (c) Salinity within CEs. (d) Temperature within CEs.

100 m within CEs [see Fig. 4(b) and (f)]. Furthermore, the R^2 (RMSE) values for the subsurface salinity inverted by $1D_{Phy}$ + 2D (21 × 21) are markedly smaller (larger) than in the case of $1D_{Phy}$ + $2D_{CBAM}$ (21 × 21), notably at around 600 m within AEs [see Fig. 4(c) and (g)] and at approximately 200 m within CEs [see Fig. 4(d) and (h)]. The depths mentioned above are speculated to correspond to regions where the background temperature and salinity fields undergo significant variations. Incorporating the CBAM in the 3D-EddyNet model allows focused attention on the temperature and salinity information at these depths, enhancing the model's inversion performance.

D. Evaluating the Impact of Eddy Physical Parameters

When removing eddy physical parameters from the 3D-EddyNet, i.e., $2D_{CBAM}$ (21 × 21), the inversion performance is the worst among the five models. Detailed information on R^2 and RMSE for the inverted subsurface temperature and salinity within eddies can be found in Table I. To further study the impact of removing eddy physical parameters on the model's performance, we conducted a comparative analysis of the residual error in subsurface temperature and salinity within eddies inverted by $1D_{Phy}$ + $2D_{CBAM}$ (21 × 21) and $2D_{CBAM}$ (21 × 21). The residual error of subsurface temperature/salinity is derived from the difference between Argo-measured and 3D-EddyNet inverted values.

As illustrated in Fig. 5, we randomly selected 100 samples from the test dataset of AEs and CEs and compared the residual errors of the above two models. The upper row in Fig. 5(a) is the subsurface temperature anomalies within AEs observed by Argo floats. Positive temperature anomalies induced by AEs exhibit varying depths, with some extending to approximately 200 m and others reaching around 600 m. The variation is attributed to the distinct oceanic regions where these eddies are located, with the former occurring in the OC region and the latter in the KE region. The middle and bottom rows in Fig. 5(a) present the residual errors of the subsurface temperature within AEs inverted by $1D_{Phy}$ + $2D_{CBAM}$ (21 × 21) and $2D_{CBAM}$ (21 × 21), respectively. The residual errors are more pronounced when using $2D_{CBAM}$ (21 × 21), with errors reaching a maximum of 2 °C. Besides, the residual errors of subsurface salinity within AEs are more pronounced when using $2D_{CBAM}$ (21 × 21), with errors reaching a maximum of 0.2 psu [see Fig. 5(b)]. Similar trends are observed for the inversion results of subsurface temperature and salinity within CEs [see Fig. 5(c) and (d)]. These results indicate that incorporating eddy physical parameters enhances model accuracy in inverting subsurface temperature and salinity within eddies.

E. Model Comparison

To thoroughly validate the accuracy and robustness of the 3D-EddyNet model, we conducted a comprehensive comparison of the 3D-EddyNet model with two contrasting methods: the eddy convolutional network (ECN) as presented by Yu et al. [50] and the residual neural network (ResNN) as proposed by Liu et al. [49]. Both ECN and ResNN were employed for retrieving subsurface temperature structures within mesoscale eddies from remote sensing data. The ECN is composed of five convolutional layers with a number of output channels: 5, 10, 10, 10, and 20 channels, followed by one FC layer. Using satellite SSHA data and Argo profiles, Yu et al. [50] employed the ECN to inverse subsurface temperature within eddies in the Northwest Pacific. On the other hand, ResNN consists of three Basic block modules, with module widths of 64, 128, and 128. Each Basic block module comprises three layers of neural networks with equal widths, along with two constant mappings. The incorporation of ResNet into Basic block modules serves to preserve input integrity and effectively mitigate the loss of input information during the propagation process. The ResNN model was applied to inverse subsurface temperature within eddies based on Argo profile and satellite SSH, and SST data.

To maintain consistency with the data used to train the 3D-EddyNet, we input seven 2-D sea surface images sized 21×21 into the ECN model. These images include latitude, longitude, SSHA, SSTA, SSSA, SSWUA, and SSWVA maps centered around eddy centers. Besides, nine 1-D parameters (eddy radius, velocity, amplitude, EKE, locations of eddy centers in latitude and longitude, the meridional and zonal distances (between Argo profiles and eddy centers, and dates of eddies) and five sea surface parameters (SSHA, SSTA, SSSA, SSWUA, and SSWVA) located nearest to Argo profiles' positions are input into the ResNN model.

The performance of the three models on the test set is illustrated in Fig. 4, depicting their R^2 (RMSE) of subsurface temperature and salinity at different depth levels within AEs and CEs. Remarkably, the R^2 values for ECN and ResNN, as validated by the test dataset, are significantly smaller, while the RMSE values are larger than those for the 3D-EddyNet model configured as $1\mathrm{D}_{\mathrm{Phy}} + 2\mathrm{D}_{\mathrm{CBAM}}$ (21×21). Furthermore, the ResNN exhibits inferior performance compared to ECN. Quantitative information on R^2 and RMSE for the three models is provided in Table I. These comparative experiments conclusively demonstrate that our 3D-EddyNet model outperforms ECN and ResNN in terms of accuracy and robustness.

V. Results

A. Subsurface Temperature and Salinity Inversion for Eddies in the KE and the OC

As depicted in Figs. 4 and 5, the 3D-EddyNet model configured as $1\mathrm{D}_{\mathrm{Phy}} + 2\mathrm{D}_{\mathrm{CBAM}}$ (21×21) demonstrates superior performance in inverting subsurface temperature and salinity within eddies. We used this model to conduct subsurface temperature and salinity inversion for eddies without Argo profiles. Considering that varying temperature and salinity background fields can lead to differences in the 3-D temperature and salinity structures within eddies, we inverted the subsurface temperature and salinity structures of eddies in the KE and the OC, respectively. We randomly selected 100 AEs and 100 CEs in the KE and OC regions, all lacking Argo profiles. Leveraging these eddies' physical parameters and corresponding surface sea images, we employed the 3D-EddyNet model to retrieve their subsurface temperature and salinity.

Fig. 6 displays the subsurface temperature and salinity within AEs and CEs in the KE region inverted by the 3D-EddyNet model. Fig. 6(a) and (b) depicts the subsurface temperature and salinity at the center of AEs inverted by the model, respectively, which both exhibit positive anomalies within the upper 600 m, with the maximum primarily concentrated between 200 and 400 m. In contrast, Fig. 6(e) and (f) represents the subsurface temperature and salinity at the center of CEs in the KE region as inferred by the model inversion, respectively. These exhibit negative anomalies within the upper 600 m, with their maxima primarily distributed from 200 to 400 m. The characteristics of subsurface temperature and salinity within eddies, as inverted by the 3D-EddyNet model, show good consistency with previously established research findings [18], [21], [70].

Given that the 3D-EddyNet model is designed to obtain the 3-D temperature and salinity structures within eddies, we not only aim to invert the subsurface temperature and salinity at the center of eddies but also possess the capability to retrieve the subsurface temperature and salinity at any point within the eddy. To illustrate this capability of the model, we inverted the subsurface temperature and salinity at the edges of AEs and CEs—specifically, at a distance equivalent to one eddy radius from the center. We then calculated the differences in the subsurface temperature and salinity between eddy edges and centers. The results reveal that the subsurface temperature and salinity at the center of AEs are significantly higher than at their edges [see Fig. 6(c) and (d)], whereas the subsurface temperature and salinity at the center of CEs are notably lower than at their edges [see Fig. 6(g) and (h)].

Fig. 7 illustrates the subsurface temperature and salinity of 100 AEs and 100 CEs in the OC inverted by the 3D-EddyNet model. As depicted in Fig. 7(a) and (b) and (e) and (f), subsurface temperature and salinity at centers of AEs (CEs) are predominantly positive (negative) in the upper 200 m. The results align with the previous research on the eddy-induced subsurface temperature and salinity structures in the OC [71], [72]. We also compared the subsurface temperature and salinity differences in eddy centers and edges in the OC region. As shown in Fig. 7(c) and (d) and (g) and (h), subsurface temperature and salinity at the center of AEs (CEs) are significantly higher (lower) than at their edges.

By comparing eddy-induced subsurface temperature and salinity differences at eddy centers and edges in the KE and OC regions, we found that subsurface temperature and salinity anomalies weaken as they move farther from the eddy center. Such a result aligns with eddies' intrinsic thermodynamic characteristics [3]. Consequently, the 3D-EddyNet model is capable of accurately inverting the temperature and salinity anomalies horizontally across various depth layers within eddies. This capability enhances the efficacy of reconstructing the 3-D temperature and salinity structures of mesoscale eddies.

B. Reconstruction of 3-D Eddy Structure in the KE and the OC

Next, we reconstruct the 3-D eddy structures by inverting subsurface temperature and salinity for each point within a

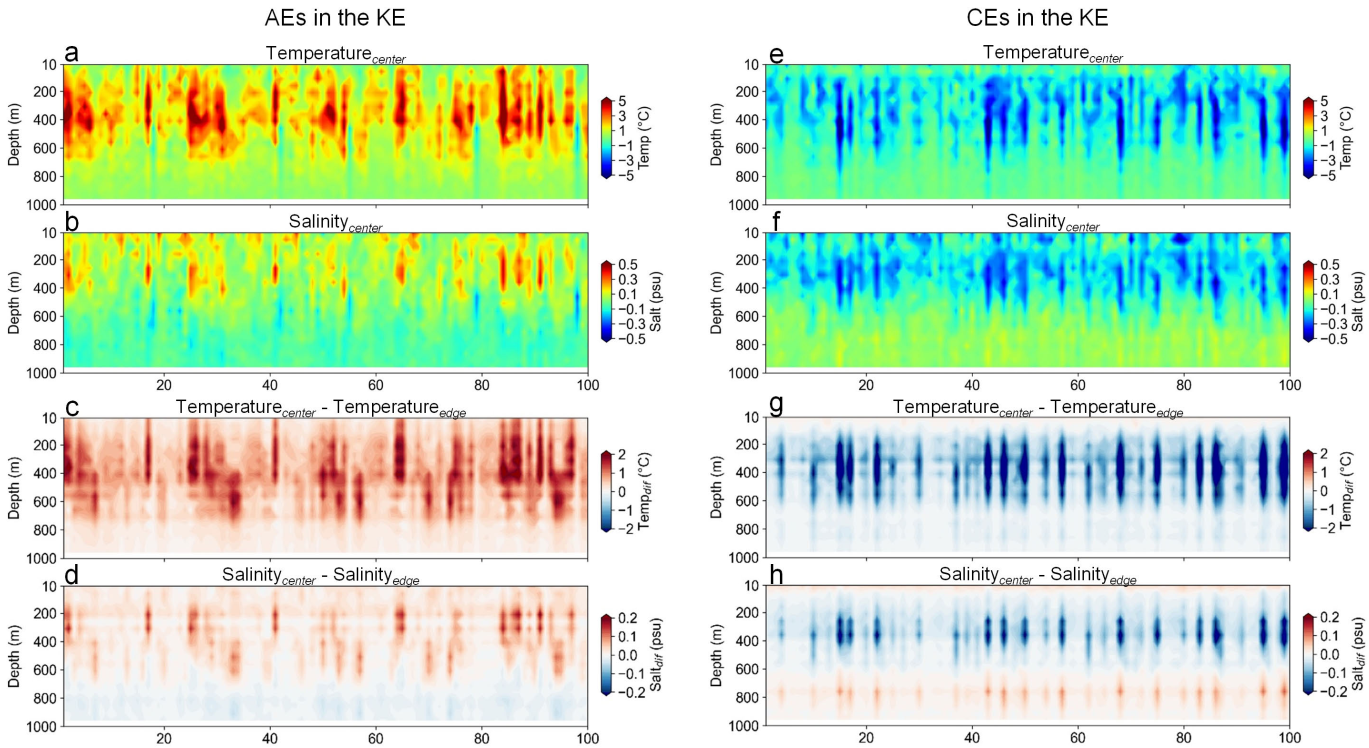

Fig. 6. 3D-EddyNet inversed subsurface temperature and salinity in eddy centers (a) and (b) and (e) and (f), and their difference from eddy edges (c) and (d) and (g) and (h) for AEs (a)–(d) and CEs (e)–(h) in the KE.

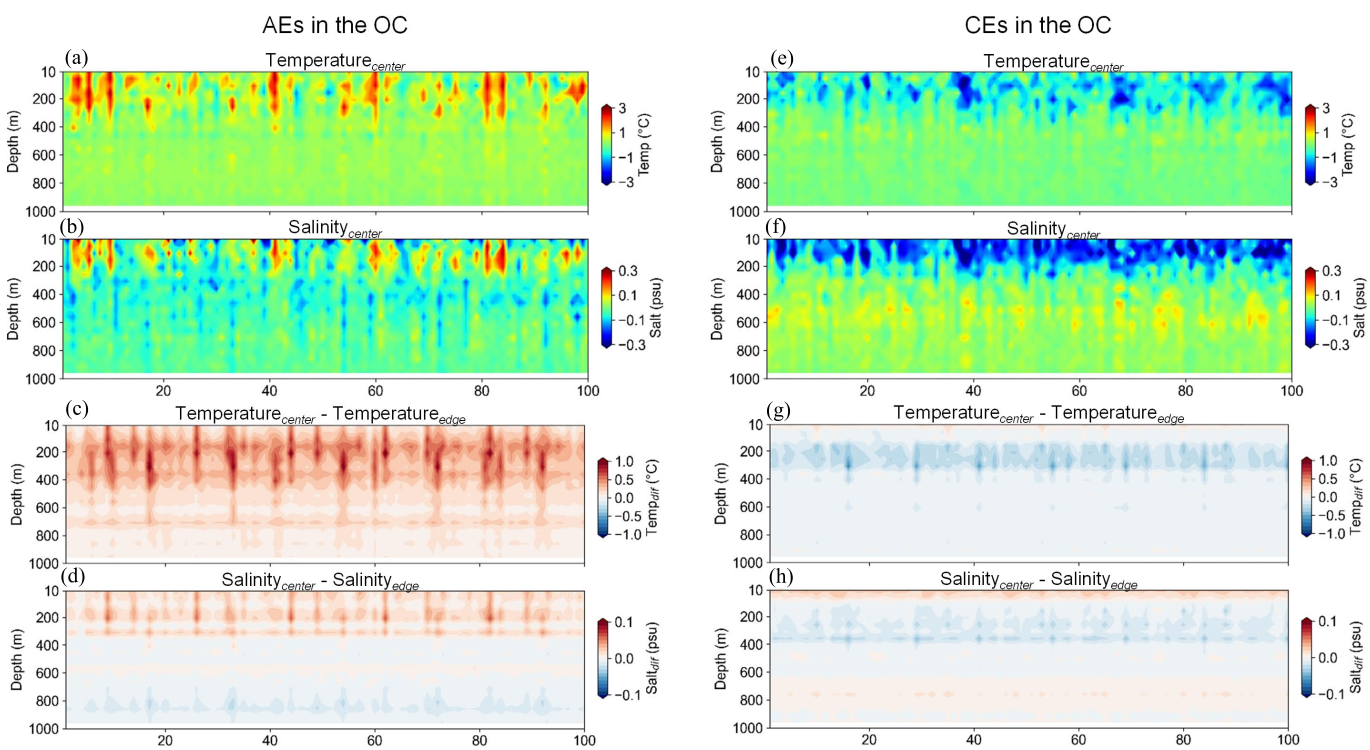

Fig. 7. 3D-EddyNet inversed subsurface temperature and salinity in eddy centers (a) and (b) and (e) and (f) and their difference from eddy edges (c) and (d) and (g) and (h) for AEs (a)–(d) and CEs (e)–(h) in the OC.

31×31 grid (i.e., $1.5R \times 1.5R$) centered on the eddy center at various depth layers using the 3D-EddyNet model. The study presents individual case studies of the 3-D temperature and salinity structures within AEs and CEs in the KE and OC regions, reconstructed using the 3D-EddyNet model. To aid visualization, temperature and salinity anomalies induced by eddies are depicted at various depths from 10 to 1000 m, with 100-m intervals.

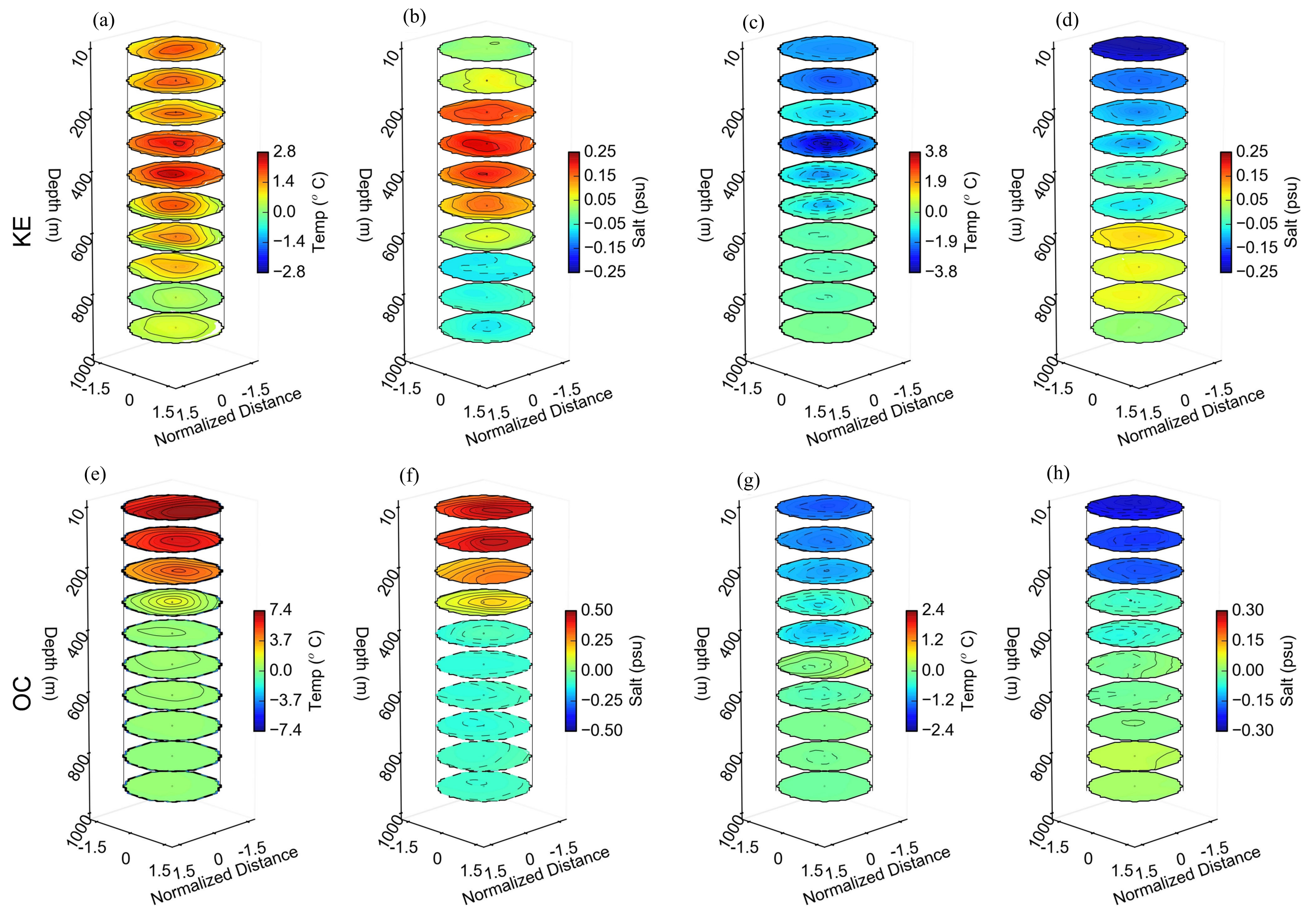

Fig. 8. Case study of reconstructed 3-D temperature and salinity structures of AEs (a), (b), (e), and (f) and CEs (c), (d), (g), and (h) in the KE (a)–(d) and the OC (e)–(h).

Fig. 8(a) and (b) illustrates the 3-D temperature and salinity structures within an AE in the KE, respectively, as reconstructed by the 3D-EddyNet model. The AE induces positive anomalies in temperature and salinity, primarily concentrated within the upper 600 m, with peak values around 300 m. Furthermore, the magnitude of temperature and salinity anomalies at each depth level is highest near the eddy center, gradually decreasing as one moves to the eddy edge. Conversely, the CE induces negative temperature and salinity anomalies [see Fig. 8(c) and (d)], predominantly distributed within the upper 500 m vertically, with the horizontal extent of the anomalies weakening as the distance from the eddy center increases.

Fig. 8(e)–(h) shows the 3-D temperature and salinity structures within an AE and a CE in the OC. The AE results in positive temperature and salinity anomalies primarily within the upper 300 m [see Fig. 8(e) and (f)], while the CE induces negative temperature and salinity anomalies extending up to the upper 400 m [see Fig. 8(g) and (h)]. Moreover, the absolute values decrease for temperature and salinity at the specific depth level within eddies as the distance from the eddy center increases.

C. Comparison Between ARMOR3D and Model-Based Eddy Structures

It is essential to acknowledge that the reconstructed 3-D eddy structures described above lack verification due to the absence of in situ data. Therefore, we rely on widely accepted 3-D reanalysis temperature and salinity data, specifically the ARMOR3D dataset, for validation. It is important to note that the ARMOR3D dataset has a temporal resolution of one week, which cannot directly validate the 3-D structures of individual eddies inversed by the 3D-EddyNet model. Therefore, we conducted a comparative analysis of the mean vertical profiles of temperature and salinity anomalies, averaged within eddies constructed by the 3D-EddyNet and the ARMOR3D dataset, respectively. In the study, we reconstructed the 3-D structures of eddies that were not associated with Argo profiles in the KE and OC regions from 2010 to 2015. Specifically, this encompassed 52 382 AEs and 55 055 CEs in the KE region, as well as 45 165 AEs and 43 680 CEs in the OC region.

The vertical temperature and salinity of eddies, as derived from the 3D-EddyNet model, exhibit a commendable agreement with corresponding data from the ARMOR3D dataset. As shown in Fig. 9(a), AEs (CEs) cause positive (negative) temperature anomalies throughout the water column, with maximum (minimum) located between 200 and 400 m, exceeding 2 °C (−2 °C). Unlike the single-core temperature structure within eddies in the KE, they exhibit a dual-core salinity structure. As illustrated in Fig. 9(b), the mean vertical salinity profiles of AEs (CEs) show positive (negative) anomalies in the upper 600 m, with the maximum (minimum) located at approximately 200 m, exceeding 0.1 psu (−0.2 psu).

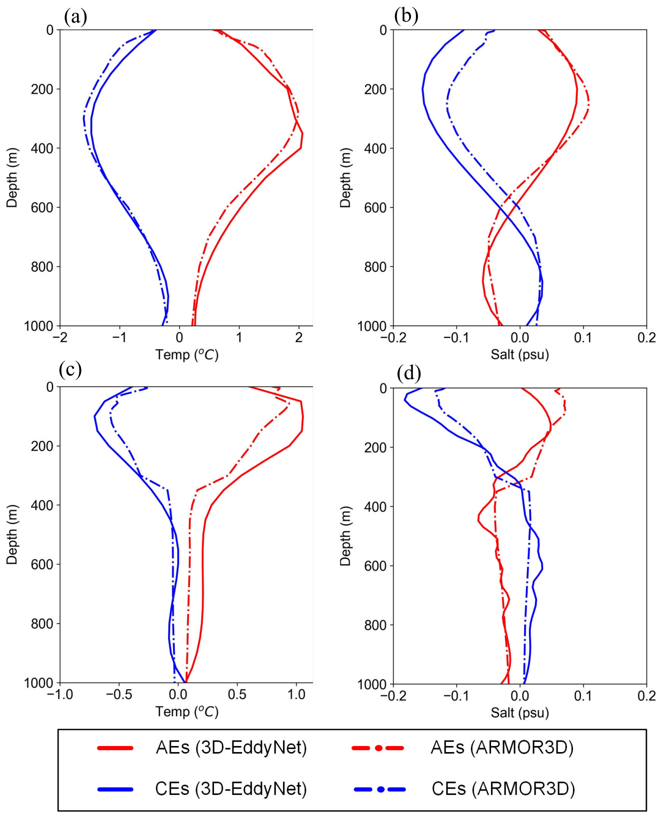

Fig. 9. Mean vertical profiles of temperature and salinity anomalies averaged within CEs (blue) and AEs (red) in the KE (a) and (b) and the OC (c) and (d) based on 3D-EddyNet (solid curves) and ARMOR3D (dashed curves).

Besides, AEs (CEs) show negative (positive) salinity anomalies below 600 m. The specific vertical salinity structures within eddies are caused by the presence of high-salinity North Pacific Tropical Water (NPTW) near the sea surface and low-salinity North Pacific Intermediate Water (NPIW) at deeper depths near the KE region [70].

Fig. 9(c) and (d) shows the mean vertical profiles of temperature and salinity anomalies averaged within AEs and CEs in the OC region, respectively. Similar to the finding in the KE, the mean vertical temperature and salinity profiles of eddies obtained from the 3D-EddyNet model are in good agreement with the ARMOR3D dataset in the OC. As shown in Fig. 9(c) AEs (CEs) show positive (negative) temperature anomalies throughout the whole depth, with the maximum (minimum) at 100 m exceeding 1 °C (−0.8 °C). Moreover, eddy-induced salinity anomalies within AEs (CEs) exhibit a dual-core structure in the OC region. As shown in Fig. 9(d), AEs (CEs) show positive (negative) salinity anomalies in the upper 300 m, contrasting with their subsurface layer. It is important to note that the magnitude of salinity anomalies induced by CEs is significantly greater than that induced by AEs in the upper 200 m. The highest (lowest) salinity anomalies induced by AEs (CEs) are about 0.1 psu (−0.2 psu) located near 100 m. Such a result is consistent with previous studies, which proposed that the cold and fresh outflow of the Okhotsk Sea induces the low-salinity structures of eddies in the OC region [73], [74], [75], [76], [77].

While subtle discrepancies exist in the reconstructed vertical temperature and salinity structures within eddies when comparing the model-derived results with ARMOR3D data, these disparities can be attributed to variations in data processing methodologies. For instance, when computing temperature and salinity anomalies induced by eddies, the 3D-EddyNet model subtracts Argo data from the climatology data (CARS2009), while the ARMOR3D dataset involves subtraction from monthly averages. Nevertheless, overall, the reconstructed vertical temperature and salinity structures within eddies, based on the model and ARMOR3D data, exhibit a commendable level of consistency, particularly in the KE and OC regions. It suggests that the model demonstrates robust generalizability in inferring the 3-D structures of eddies even in the absence of Argo profiles. Furthermore, in contrast to ARMOR3D, which provides only weekly data resolution, the 3D-EddyNet model proposed in this study is capable of reconstructing 3-D eddy structures at any given time and location. This capability overcomes the limitations imposed by sparse observational data, enhancing our understanding of eddy dynamics and their impact on oceanic processes.

VI. Discussion and Conclusion

In this study, we present 3D-EddyNet, a DL-based model specifically tailored for reconstructing the 3-D thermohaline structure of mesoscale eddies. Deviating from traditional DL-based models relying solely on sea surface remote sensing data for inferring subsurface temperature and salinity [43], [44], [49], [50], our study integrates prior knowledge of eddies through the construction of a dual-branch network, resulting in a significant improvement in inversion performance. Notably, compared to previous DL-based models, the 3D-EddyNet exhibits a substantial enhancement in R^2 values ranging from 10% to 20%, accompanied by a reduction in RMSE ranging from 20% to 40%. Furthermore, our investigation unveils a noteworthy insight regarding the dimensions of input 2-D images. Unlike previous studies focusing on large-scale ocean basins, increasing the dimensions of sea surface images does not boost the inversion performance of the 3D-EddyNet model when targeting mesoscale eddies. Optimal performance is achieved when using sea surface images with dimensions close to the average size of mesoscale eddies in the study area. This observation emphasizes the intricate nature of the targets for inversion, where larger sea surface images may not necessarily provide additional information relevant to mesoscale eddies. In addition, the incorporation of the CBAM module in the 3D-EddyNet model further enhances inversion performance, particularly when estimating subsurface temperature within AEs (CEs) at depths of approximately 200 m (100 m), as well as subsurface salinity within AEs (CEs) at around 600 m (200 m). The result emphasizes the significance of utilizing advanced architectures for enhancing model accuracy.

Based on the optimized 3D-EddyNet model, we conducted a comprehensive analysis of eddies that were not associated with Argo profiles in the KE and OC regions. The characteristics of subsurface temperature and salinity structures inverted by the 3D-EddyNet model in the KE and the OC regions are consistent with previous research findings. Furthermore, the

3D-EddyNet not only exhibits robust capabilities in accurately representing the depth-dependent variations of subsurface temperature and salinity within eddies, and the 3D-EddyNet can also capture the variations in temperature and salinity with respect to the distance from the eddy center in the horizontal direction. As a result, the model's ability to reconstruct 3-D eddy structures demonstrates excellent generalization, validated through comparison with the weekly ARMOR3D data. The real-time reconstruction capacity of the 3D-EddyNet model, overcoming the constraints posed by the weekly data resolution, not only provides valuable insights into oceanic dynamics but also enhances our ability to investigate eddy-related phenomena. While the primary purpose of the model design is to invert the subsurface thermohaline structures of mesoscale eddies, the model can also be applied to the subsurface thermohaline inversion of large-scale ocean basins. Adjusting parameters at the data input stage, such as using sea surface images centered on Argo profile positions instead of eddy centers, enhances the model's applicability to various oceanic scales.

In summary, the study provides an effective tool for conducting high spatiotemporal resolution research on the 3-D thermohaline structure of mesoscale eddies. The model helps to accurately assess the role of mesoscale eddies in transporting mass and energy in the global ocean. In addition, it offers an efficient approach for data reconstruction and parameter inversion of various marine phenomena. Despite achieving success in reconstructing 3-D eddy structures, the 3D-EddyNet model is not without its limitations and uncertainties. Concerning data input, the relatively low spatiotemporal resolution of SSS data may introduce errors through interpolation, affecting the accuracy of inversion results. Future research requires higher resolution SSS data or more precise interpolation methods to enhance the model's reliability. In terms of model's design, incorporating eddy prior knowledge solely at the input stage may not be sufficient to comprehensively describe the thermohaline characteristics of mesoscale eddies. Beyond the data input stage, there are several proposed methods to improve model accuracy and reliability by integrating domain knowledge with neural networks, including constructing learning biases through loss functions, utilizing specific model structures, forming hard constraints based on special activation functions, and combinations of these schemes [78], [79], [80], [81]. However, the exploration of AI-based models with physical constraints in the marine domain is in its early stages, requiring further investigation to incorporate these constraints at various model development stages for improved inversion accuracy of 3-D eddy structures.

ACKNOWLEDGMENT

The SST data can be downloaded from National Oceanic Atmospheric Administration (NOAA, https://www.ncei.noaa.gov/products/optimum-interpolation-sst). The SSH data are available on the website of Copernicus Marine and Environmental Monitoring Service (CMEMS, https://doi.org/10.48670/moi-00148). The SSS data are available at CMEMS (https://doi.org/10.48670/moi-00051). The SSW data can be downloaded from the Remote Sensing Systems (RSS, https://www.remss.com/measurements/ccmp/. The mesoscale eddies dataset can be downloaded from https://figshare.com/s/3c3b03776d9862ac85bc. Argo data can be downloaded from https://data-argo.ifremer.fr/dac/. CARS2009 data can be downloaded from http://www.marine.csiro.au/~dunn/cars2009/. The ARMOR3D data can be downloaded from https://doi.org/10.48670/moi-00052.

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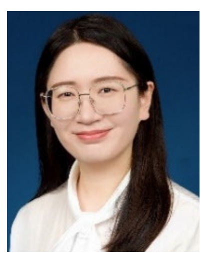

Yingjie Liu (Member, IEEE) received the B.S. degree in resources environment and the management of urban and rural planning from Qingdao University of Technology, Qingdao, China, in 2013, the Ph.D. degree in geography information system from the Ocean University of China, Qingdao, in 2019, and the joint Ph.D. degree in physical oceanography from the Woods Hole Oceanographic Institution, Falmouth, MA, USA, in 2019.

In 2019, she joined the Institute of Oceanology, Chinese Academy of Sciences, Qingdao, China, where she is currently an Assistant Researcher. Her research interests include satellite oceanography and the application of artificial intelligence in mesoscale eddies.

Haoyu Wang (Member, IEEE) is currently pursuing the Ph.D. degree in physical oceanography with the Institute of Oceanology, Chinese Academy of Sciences, Qingdao, China.

He is also a Visiting Graduate Student at Duke University, Durham, NC, USA. His research interests include El Niño/Southern Oscillation (ENSO) forecasting and estimation of oceanic subsurface variables through the application of remote sensing and artificial intelligence techniques.

Fei Jiang (Graduate Student Member, IEEE) received the B.S. degree in ocean technology from the Ocean University of China, Qingdao, China, in 2022. She is currently pursuing the joint M.S. degree in physical oceanography with the Institute of Oceanology, Chinese Academy of Sciences, Qingdao, and the University of Chinese Academy of Sciences, Beijing, China.

Her research interests include satellite oceanography and the study of ocean eddies using deep learning techniques.

Yuan Zhou (Senior Member, IEEE) received the B.Eng., M.Eng., and Ph.D. degrees in electronic engineering and communication engineering from Tianjin University, Tianjin, China, in 2006, 2008, and 2011, respectively.

Since 2011, she has been a Faculty Member with the School of Electronic Information Engineering, Tianjin University, where she is currently a Professor. From 2013 to 2014, she was a Visiting Scholar with the School of Mechanical and Electrical Engineering, University of Southern Queensland, Toowoomba, QLD, Australia. From 2016 to 2017, she was a Visiting Scholar with the Department of Electrical Engineering, Princeton University, Princeton, NJ, USA. Her research interests include computer vision, machine learning, intelligent interactive robot, and artificial intelligence oceanography.

Xiaofeng Li (Fellow, IEEE) received the bachelor's degree in optical engineering from Zhejiang University, Hangzhou, China, in 1985, the master's degree in physical oceanography from the First Institute of Oceanography, Qingdao, China, in 1992, and the Ph.D. degree in physical oceanography from North Carolina State University, Raleigh, NC, USA, in 1997.

He was with the National Environmental Satellite, Data, and Information Service (NESDIS), National Ocean and Atmospheric Administration (NOAA), WA, USA, from 1997 to 2019, where he was involved in developing many operational satellite ocean remote sensing products. He is with the Institute of Oceanology, Chinese Academy of Sciences, Qingdao. His research interests include synthetic aperture radar (SAR) applications in oceanography and marine meteorology, the application of artificial intelligence in oceanographic studies, big data analytics, and satellite image processing.

Dr. Li holds editorial positions as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *International Journal of Remote Sensing*, an Editor for *JGR: Machine Learning and Computation*, and the Executive Editor-in-Chief for the *Journal of Remote Sensing*.